

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250281987

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

KUBODERA; Yu

METHOD AND SYSTEM FOR GRINDING WORKPIECE

Abstract

A method for grinding a workpiece includes a first grinding step in which a gear-shaped workpiece and a grinding tool are meshed and rotated, whereby a workpiece tooth surface is ground with a grinding tooth surface, a first correction information generating step in which first correction information is generated by inverting a first deflection waveform, and a second grinding step in which in a state where the workpiece and the grinding tool are meshed and rotated, a rotational speed of the workpiece or the grinding tool is changed based on the first correction information, whereby the workpiece tooth surface is ground with the grinding tooth surface.

Inventors: KUBODERA; Yu (TOKYO, JP)

Applicant: HONDA MOTOR CO., LTD. (TOKYO, JP)

Family ID: 1000008507891

Appl. No.: 19/058202

Filed: February 20, 2025

Foreign Application Priority Data

JP	2024-036952	Mar. 11, 2024
----	-------------	---------------

Publication Classification

Int. Cl.: B23F5/04 (20060101); B23F21/02 (20060101)

U.S. Cl.:

CPC B23F5/04 (20130101); B23F21/026 (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2024-036952 filed on Mar. 11, 2024, the contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

Field of the Invention

[0002] The present disclosure relates to a method and a system for grinding a workpiece.

Description of the Related Art

[0003] JP 5467833 B2 discloses a method for grinding a workpiece in which a gear-shaped workpiece and a grinding tool are meshed and rotated to grind a workpiece tooth surface of the workpiece with a helical grinding tooth surface of the grinding tool.

SUMMARY OF THE INVENTION

[0004] A better method and system for grinding a workpiece are long-awaited.

[0005] The present disclosure aims to solve the aforementioned problems.

[0006] A first aspect of the present disclosure is a method for grinding a workpiece, comprising: a first grinding step in which a gear-shaped workpiece and a grinding tool are meshed and rotated, whereby a workpiece tooth surface is ground with a helical grinding tooth surface of the grinding tool; a first correction information generating step in which first correction information is generated by inverting a first deflection waveform indicating a relationship between a rotational phase of the workpiece and a positional deviation amount of the workpiece tooth surface with respect to the grinding tooth surface during the first grinding step; and a second grinding step in which in a state where the workpiece and the grinding tool are meshed and rotated, a rotational speed of the workpiece or the grinding tool is changed based on the first correction information, whereby the workpiece tooth surface is ground with the grinding tooth surface.

[0007] A second aspect of the present disclosure is a grinding system comprising: a grinding control unit configured to perform a first grinding step in which a gear-shaped workpiece and a grinding tool are meshed and rotated, whereby a workpiece tooth surface of the workpiece is ground with a helical grinding tooth surface of the grinding tool; and a correction information generation unit configured to generate first correction information by inverting a first deflection waveform that indicates a relationship between a rotational phase of the workpiece and a positional deviation amount of the workpiece tooth surface with respect to the grinding tooth surface during the first grinding step, wherein the grinding control unit further performs a second grinding step in which in a state where the workpiece and the grinding tool are meshed and rotated, a rotational speed of the workpiece or the grinding tool is changed based on the first correction information, whereby the workpiece tooth surface is ground with the grinding tooth surface.

[0008] According to the present disclosure, a better method and system for grinding a workpiece are provided.

[0009] The above and other objects, features, and advantages of the present invention will become more apparent from the following description when taken in conjunction with the accompanying drawings, in which a preferred embodiment of the present invention is shown by way of illustrative example.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. 1 is a perspective view of a grinding system according to an embodiment;

[0011] FIG. 2 is a control block diagram for the grinding system;

[0012] FIG. 3 is a flowchart showing an example of a method for grinding a workpiece;

[0013] FIG. 4 is an explanatory diagram for a first correction information generation step; [0014] FIG. 5 is an explanatory diagram for a second correction information generation step; and [0015] FIG. 6 is a graph showing a deflection waveform of a third grinding step.

DETAILED DESCRIPTION OF THE INVENTION

[0016] Gear-shaped workpieces are subjected to heat treatment before the grinding. In this case, a workpiece may deform into an elliptical shape due to thermal strain when viewed from the direction of the rotational axis of the workpiece. Then, there is a possibility that a workpiece tooth surface cannot be ground with good accuracy by a grinding tooth surface. The present disclosure can provide a method and a system for grinding a workpiece that can accurately grind a workpiece tooth surface.

[0017] FIG. 1 is a perspective view of a grinding system **10** according to an embodiment. As shown in FIG. 1, the grinding system **10** is a system for grinding a gear-shaped workpiece **12** with a grinding tool **14**. The grinding system **10** includes a bed **16**, a gear support mechanism **18**, a gear rotation mechanism **20**, a tool support mechanism **22**, a tool rotation mechanism **24**, and a controller **26**.

[0018] The bed **16** is placed, for example, on a horizontal surface of a factory or the like. The gear support mechanism **18** is disposed on a flat upper surface of the bed **16**. The gear support mechanism **18** includes a cutting table **28**, a cutting motor **30**, a traverse table **32**, and a traverse motor **34**.

[0019] The cutting table **28** moves in the A direction with respect to the bed **16**. The A direction is a horizontal direction perpendicular to the height direction of the bed **16**. The cutting table **28** is connected to the cutting motor **30** via a ball screw shaft **36**. The cutting motor **30** moves the cutting table **28** in the A direction by rotating the ball screw shaft **36**.

[0020] The traverse table **32** is disposed on an upper surface of the cutting table **28**. The traverse table **32** moves in the B direction with respect to the cutting table **28**. The B direction is a direction perpendicular to the height direction of the bed **16** and the A direction. The traverse table **32** is coupled to the traverse motor **34** via a ball screw shaft (not shown). The traverse motor **34** moves the traverse table **32** in the B direction by rotating the ball screw shaft.

[0021] The gear rotation mechanism **20** is arranged on an upper surface of the traverse table **32**. The gear rotation mechanism **20** has a gear mounting shaft **38** and a first motor **40**. The gear mounting shaft **38** extends in the B direction. The workpiece **12** is attachable to and detachable from the gear mounting shaft **38**. The first motor **40** rotates the gear mounting shaft **38**.

[0022] The tool support mechanism **22** includes a column **42**, a pivot table **44**, a shift table **46**, and a shift motor **48**. The column **42** is positioned on the upper surface of the bed **16** so as to face the gear support mechanism **18**. The column **42** extends upward from the bed **16**. The pivot table **44** is attached to a surface of the column **42** facing the gear support mechanism **18**.

[0023] The pivot table **44** extends in one direction. A turning motor (not shown) turns the pivot table **44** in the C direction with respect to the column **42**. The shift table **46** is provided on a surface of the pivot table **44** facing the gear support mechanism **18**. The shift table **46** is coupled to the shift motor **48** via a ball screw shaft **50**. The shift motor **48** is attached to the pivot table **44**. The shift motor **48** moves the shift table **46** in the D direction with respect to the pivot table **44**.

[0024] The tool rotation mechanism **24** includes a base **54**, a tool mounting shaft **56**, and a second motor **58**. The base **54** is attached to a surface of the shift table **46** facing the gear support mechanism **18**. The base **54** extends along the direction in which the pivot table **44** extends. The tool mounting shaft **56** is inserted through the base **54** along the direction in which the base **54** extends. The grinding tool **14** is attachable to and detachable from the tool mounting shaft **56**. The second motor **58** rotates the tool mounting shaft **56**.

[0025] As shown in FIG. 2, the workpiece **12** is mounted on the gear mounting shaft **38**. The workpiece **12** can be rotated in the R1 direction and the R2 direction by the driving force of the first motor **40**. The workpiece **12** has a plurality of teeth **60**. Each of the teeth **60** is formed with a

workpiece tooth surface **62**. The workpiece tooth surface **62** includes a left workpiece tooth surface **62a** and a right workpiece tooth surface **62b**.

[0026] The grinding tool **14** is mounted on the tool mounting shaft **56**. The grinding tool **14** can be rotated in the R3 and R4 directions by the driving force of the second motor **58**. The grinding tool **14** is a tool for grinding the workpiece **12**. The grinding tool **14** has helical grinding teeth **64**. A grinding tooth surface **66** is formed on the grinding teeth **64**. The grinding tooth surface **66** includes a first grinding tooth surface **66a** and a second grinding tooth surface **66b**. For example, single-layer CBN (cubic boron nitride) abrasive grains or the like are electrodeposited on the grinding tooth surface **66** via a nickel plating layer.

[0027] When the workpiece **12** is ground by the grinding tool **14**, the workpiece **12** and the grinding tool **14** are meshed with each other. With the workpiece **12** and the grinding tool **14** meshed, the left workpiece tooth surface **62a** faces the first grinding tooth surface **66a**, and the right workpiece tooth surface **62b** faces the second grinding tooth surface **66b**. With the workpiece **12** and the grinding tool **14** meshed with each other, the workpiece **12** is rotated in the R1 direction and the grinding tool **14** is rotated in the R3 direction, for example, whereby the left workpiece tooth surface **62a** can be ground by the first grinding tooth surface **66a** and the right workpiece tooth surface **62b** can be ground by the second grinding tooth surface **66b**. With the workpiece **12** and the grinding tool **14** meshed with each other, the workpiece **12** is rotated in the R2 direction and the grinding tool **14** is rotated in the R4 direction, for example, whereby also the left workpiece tooth surface **62a** can be ground by the first grinding tooth surface **66a** and the right workpiece tooth surface **62b** can be ground by the second grinding tooth surface **66b**.

[0028] The grinding system **10** further includes a first encoder **68** and a second encoder **70**. The first motor **40** is provided with the first encoder **68**. The first encoder **68** outputs to the controller **26** information (e.g., pulse signals) on the rotational phase (rotational speed, rotational angle, rotational position, rotational amount) of the workpiece **12**.

[0029] The second motor **58** is provided with the second encoder **70**. The second encoder **70** outputs to the controller **26** information (e.g., pulse signals) on the rotational phase (rotational speed, rotational angle, rotational position, and rotational amount) of the grinding tool **14**.

[0030] The controller **26** includes a first servo amplifier **74**, a second servo amplifier **76**, and a control main body **78**. The first servo amplifier **74** controls the rotation of the first motor **40** based on signals output from the control main body **78**. The second servo amplifier **76** controls the rotation of the second motor **58** based on signals output from the control main body **78**.

[0031] The control main body **78** includes a computing unit **80**, a storage unit **82**, an operation unit **84**, and a display unit **86**. The computing unit **80** is composed of a processor such as a CPU (Central Processing Unit) or a GPU (Graphics Processing Unit). That is, the computing unit **80** is formed by processing circuitry.

[0032] The computing unit **80** includes a control unit **88**, a grinding control unit **90**, an information acquisition unit **92**, and a correction information generation unit **94**. The control unit **88** controls the cutting motor **30**, the traverse motor **34**, the turning motor (not shown), and the shift motor **48**. The grinding control unit **90** controls the rotation of the workpiece **12** via the first servo amplifier **74**. The grinding control unit **90** controls the rotation of the grinding tool **14** via the second servo amplifier **76**. The information acquisition unit **92** acquires information output from the first encoder **68** and information output from the second encoder **70**. The correction information generation unit **94** generates first correction information **104** and second correction information **110**, which will be described later.

[0033] The control unit **88**, grinding control unit **90**, information acquisition unit **92**, and correction information generation unit **94** can be realized by the computing unit **80** executing programs stored in the storage unit **82**. At least part of the control unit **88**, the grinding control unit **90**, the information acquisition unit **92**, and the correction information generation unit **94** may be realized by an integrated circuit such as an ASIC (Application Specific Integrated Circuit), an FPGA (Field-

Programmable Gate Array), or the like. In addition, at least part of the control unit **88**, the grinding control unit **90**, the information acquisition unit **92**, and the correction information generation unit **94** may be configured by an electronic circuit including discrete devices.

[0034] The storage unit **82** is composed of a volatile memory (not shown) and a nonvolatile memory (not shown). Examples of the volatile memory include, for example, a RAM (Random Access Memory) or the like. The volatile memory is used as working memory of a processor to temporarily store data or the like required for processing or computing operations. Examples of the nonvolatile memory include, for example, a ROM (Read Only Memory), a flash memory, or the like. The non-volatile memory is used as memory for storage, storing programs, tables, maps, etc. At least part of the storage unit **82** may be provided in the above-described processor, integrated circuit, etc.

[0035] The operation unit **84** is used when a user operates the controller **26**. The operation unit **84** may include a keyboard, a mouse, and the like. The display unit **86** is provided with a display element (not shown). As the display element, for example, a liquid crystal display element, an organic electroluminescence display element, or the like is used. The operation unit **84** and the display unit **86** may be configured by a touch panel (not shown) provided with such a display element.

[0036] Next, an example of a method for grinding the workpiece **12** will be described. FIG. **3** is a flowchart showing an example of a method for grinding a workpiece **12**. The workpiece **12** is subjected to heat treatment before the grinding. In this case, the workpiece **12** may deform into an elliptical shape due to thermal strain when viewed from the direction of the rotational axis of the workpiece **12**.

[0037] In step **S1**, the workpiece **12** is mounted on the gear mounting shaft **38** and the grinding tool **14** is mounted on the tool mounting shaft **56**. Thereafter, the process transitions to step **S2**.

[0038] In step **S2**, the workpiece **12** and the grinding tool **14** are meshed with each other. Specifically, the control unit **88** controls the cutting motor **30**, the traverse motor **34**, the turning motor (not shown), and the shift motor **48** to mesh the workpiece **12** with the grinding tool **14**. Thereafter, the process transitions to step **S3**.

[0039] In step **S3**, a first grinding step is performed. In the first grinding step, the workpiece tooth surface **62** is subjected to rough grinding (first rough grinding). The grinding control unit **90** rotates the workpiece **12** via the first servo amplifier **74** and rotates the grinding tool **14** via the second servo amplifier **76**. The grinding control unit **90** rotates the workpiece **12** and the grinding tool **14** synchronously. In other words, the grinding control unit **90** controls the first servo amplifier **74** and the second servo amplifier **76** using feedback in a manner so that the workpiece **12** and the grinding tool **14** rotate maintaining the engaged state, based on the information output from the first encoder **68** and the information output from the second encoder **70**.

[0040] In the first grinding step, grinding is performed over the circumference of the workpiece **12** (on all the workpiece tooth surfaces **62**). If thermal strain has caused deformation in the workpiece **12**, deflection of the workpiece tooth surface **62** with respect to the grinding tooth surface **66** can occur in the first grinding step. Thereafter, the process transitions to step **S4**.

[0041] In step **S4**, a first correction information generation step is performed. FIG. **4** is an explanatory diagram for a first correction information generation step. In FIG. **4**, the horizontal axis indicates the rotational phase of the workpiece **12** and the vertical axis indicates the rotational phase difference of the workpiece **12**. In the first correction information generation step, the difference (rotational phase difference) between a command value of the rotational phase of the workpiece **12** during the first grinding step and a detected value of the rotational phase of the workpiece **12** is acquired as a positional deviation amount of the workpiece tooth surface **62** with respect to the grinding tooth surface **66**. The rotational phase difference of the workpiece **12** indicates a synchronization error in the rotation of the workpiece **12**.

[0042] As shown in FIG. **4**, in the first correction information generation step, the correction

information generation unit **94** generates a first deflection waveform **102** by approximating with a sinusoidal wave the deflection waveform **100** that indicates the relationship between the amount of deflection of the workpiece tooth surface **62** with respect to the grinding tooth surface **66** and the rotational phase of the workpiece **12** during the first grinding step. That is, the first deflection waveform **102** indicates the changes in the positional deviation amount of the workpiece tooth surface **62** over one complete lap around the workpiece **12** in the first grinding step. In other words, the first deflection waveform **102** indicates a grinding error in the first grinding step. In the first correction information generation step, the correction information generation unit **94** generates the first correction information **104** by inverting the first deflection waveform **102**. The first correction information **104** is information indicating a waveform obtained by inverting the first deflection waveform **102** in the direction of the vertical axis of FIG. 4 (the deflection width direction of the first deflection waveform **102**). Thereafter, the process transitions to step S5.

[0043] In step S5, a second grinding step is performed. In the second grinding step, the workpiece tooth surface **62** is subjected to rough grinding (second rough grinding). In the second grinding step, while the workpiece **12** and the grinding tool **14** are meshed and rotated, the grinding control unit **90** changes the rotational speed of the workpiece **12** based on the first correction information **104**, thereby grinding the workpiece tooth surface **62** with the grinding tooth surface **66**. In other words, the grinding control unit **90** outputs to the first servo amplifier **74** the command signal generated based on the first correction information **104** and a synchronous rotation signal of the workpiece **12** and also outputs a synchronous rotation signal of the grinding tool **14** to the second servo amplifier **76**, thereby grinding the workpiece tooth surface **62** with the grinding tooth surface **66**. That is, in the second grinding step, the grinding control unit **90** controls the rotational speed of the workpiece **12** using feedback so that the grinding error generated in the first grinding step becomes small (a deflection component is corrected).

[0044] Thus, in the second grinding step, the grinding error generated in the first grinding step can be reduced. In the second grinding step, in a state where the workpiece **12** and the grinding tool **14** are meshed and rotated, the workpiece tooth surface **62** may be ground with the grinding tooth surface **66** by changing the rotational speed of the grinding tool **14** based on the first correction information **104**. Thereafter, the process transitions to step S6.

[0045] In step S6, a second correction information generation step is performed. FIG. 5 is an explanatory diagram for a second correction information generation step. In FIG. 5, the horizontal axis indicates the rotational phase of the workpiece **12** and the vertical axis indicates the rotational phase difference of the workpiece **12**.

[0046] As shown in FIG. 5, in the second correction information generation step, the correction information generation unit **94** generates a second deflection waveform **108** by approximating with a sinusoidal wave the deflection waveform **106** that indicates the relationship between the amount of deflection of the workpiece tooth surface **62** with respect to the grinding tooth surface **66** and the rotational phase of the workpiece **12** during the second grinding step. That is, the second deflection waveform **108** indicates the changes in the positional deviation amount of the workpiece tooth surface **62** of the workpiece **12** over one complete lap around the workpiece **12** in the second grinding step. In other words, the second deflection waveform **108** indicates a grinding error in the second grinding step. In the second correction information generation step, the correction information generation unit **94** generates the second correction information **110** by inverting the second deflection waveform **108**. The second correction information **110** is information indicating a waveform obtained by inverting the second deflection waveform **108** in the direction of the vertical axis of FIG. 5 (the deflection width direction of the second deflection waveform **108**). Thereafter, the process transitions to step S7.

[0047] In step S7, a third grinding step is performed. In the third grinding step, the workpiece tooth surface **62** is subjected to finish grinding. In the third grinding step, while the workpiece **12** and the grinding tool **14** are meshed and rotated, the grinding control unit **90** changes the rotational speed

of the workpiece **12** based on the second correction information **110**, thereby grinding the workpiece tooth surface **62** with the grinding tooth surface **66**. In other words, the grinding control unit **90** outputs to the first servo amplifier **74** the command signal generated based on the second correction information **110** and the synchronous rotation signal of the workpiece **12** and also outputs a synchronous rotation signal of the grinding tool **14** to the second servo amplifier **76**, thereby grinding the workpiece tooth surface **62** with the grinding tooth surface **66**. That is, in the third grinding step, the grinding control unit **90** controls the rotational speed of the workpiece **12** using feedback so that the grinding error generated in the second grinding step becomes small (a deflection component is corrected).

[0048] FIG. **6** is a graph showing a deflection waveform **112** of the third grinding step. As shown in FIG. **6**, in the third grinding step, the grinding error generated in the second grinding step can be reduced, so that the workpiece tooth surface **62** can be ground with higher accuracy. In the third grinding step, in a state where the workpiece **12** and the grinding tool **14** are meshed and rotated, the workpiece tooth surface **62** may be ground with the grinding tooth surface **66** by changing the rotational speed of the grinding tool **14** based on the second correction information **110**. After this, the process of FIG. **3** is completed.

[0049] According to the present embodiment, the second grinding step is performed based on the first correction information **104** generated by inverting the first deflection waveform **102** that indicates the relationship between the positional deviation amount of the workpiece tooth surface **62** and the rotational phase of the workpiece **12** during the first grinding step. Thus, in the second grinding step, the grinding error generated in the first grinding step can be reduced. Therefore, the workpiece tooth surface **62** can be ground with high accuracy. Accordingly, a better method and system **10** for grinding the workpiece **12** can be provided.

[0050] With respect to the above embodiments, the following supplementary notes are further disclosed.

Supplementary Note 1

[0051] A method for grinding a workpiece of the present disclosure includes: performing first grinding in which a gear-shaped workpiece (**12**) and a grinding tool (**14**) are meshed and rotated, whereby a workpiece tooth surface (**62**) of the workpiece is ground with a helical grinding tooth surface (**66**) of the grinding tool; generating first correction information (**104**) by inverting a first deflection waveform (**102**) that indicates a relationship between a rotational phase of the workpiece and a positional deviation amount of the workpiece tooth surface with respect to the grinding tooth surface during the first grinding; and performing a second grinding in which in a state where the workpiece and the grinding tool are meshed and rotated, a rotational speed of the workpiece or the grinding tool is changed based on the first correction information, whereby the workpiece tooth surface is ground with the grinding tooth surface.

[0052] According to such a configuration, the second grinding step is performed based on the first correction information generated by inverting the first deflection waveform indicating the relationship between the positional deviation amount of the workpiece tooth surface and the rotational phase of the workpiece during the first grinding step. Thus, in the second grinding step, the grinding error generated in the first grinding step can be reduced. Therefore, the workpiece tooth surface can be ground with high accuracy. Therefore, a better method for grinding the workpiece can be provided.

Supplementary Note 2

[0053] The method for grinding the workpiece according to Supplementary note 1 may acquire, as the positional deviation amount, a difference between a command value of a rotational phase of the workpiece and a detected value of a rotational phase of the workpiece during the first grinding step in the first correction information generation step.

[0054] According to such a configuration, the first correction information can be generated easily.

Supplementary Note 3

[0055] The method for grinding the workpiece according to Supplementary note 1 or 2 may include a second correction information generating step for generating second correction information (**110**) by inverting a second deflection waveform (**108**) indicating the relationship between the rotational phase of the workpiece and the positional deviation amount of the workpiece tooth surface with respect to the grinding tooth surface during the second grinding step; and a third grinding step for performing third grinding in which the workpiece tooth surface is ground with the grinding tooth surface by changing the rotational speed of the workpiece or the grinding tool based on the second correction information in a state where the workpiece and the grinding tool are meshed and rotated.

[0056] According to such a configuration, the third grinding step is performed based on the second correction information generated by inverting the second deflection waveform that indicates the relationship between the positional deviation amount of the workpiece tooth surface and the rotational phase of the workpiece during the second grinding step. Thus, in the third grinding step, the grinding error generated in the second grinding step can be reduced. Therefore, the grinding tooth surface can be ground with more accuracy.

Supplementary Note 4

[0057] A grinding system (**10**) of the present disclosure includes a grinding control unit (**90**) configured to perform a first grinding step in which a gear-shaped workpiece and a grinding tool are meshed and rotated, whereby a workpiece tooth surface of the workpiece is ground with a helical grinding tooth surface of the grinding tool, and a correction information generation unit (**94**) configured to generate first correction information by inverting a first deflection waveform indicating the relationship between the positional deviation amount of the workpiece tooth surface with respect to the grinding tooth surface and the rotational phase of the workpiece during the first grinding step, wherein the grinding control unit further performs a second grinding step in which the workpiece tooth surface is ground with the grinding tooth surface by changing the rotational speed of the workpiece or the grinding tool based on the first correction information in a state in which the workpiece and the grinding tool are meshed and rotated.

[0058] Such a configuration provides the same effect as that of Supplementary note 1. Therefore, it can provide a better grinding system.

Supplementary Note 5

[0059] In the grinding system according to Supplementary note 4, the correction information generation unit may acquire, as the positional deviation amount, a difference between a command value of a rotational phase of the workpiece and a detected value of a rotational phase of the workpiece during the first grinding step.

[0060] Such a configuration provides the same effect as that of Supplementary note 2.

Supplementary Note 6

[0061] In the grinding system according to supplementary note 4 or 5, the correction information generation unit may generate the second correction information by inverting a second deflection waveform indicating a relationship between the positional deviation amount of the workpiece tooth surface with respect to the grinding tooth surface and the rotational phase of the workpiece during the second grinding step, and the grinding control unit may further perform a third grinding step of grinding the workpiece tooth surface with the grinding tooth surface by changing the rotational speed of the workpiece or the grinding tool based on the second correction information in a state where the workpiece and the grinding tool are meshed and rotated.

[0062] Such a configuration provides the same effect as that of Supplementary note 3.

[0063] Although the present disclosure has been detailed, the present disclosure is not limited to the individual embodiments described above. These embodiments may be variously added, replaced, altered, partially deleted, etc., without departing from the scope of the present disclosure or the intent of the present disclosure as derived from the claims and their equivalents. These embodiments can also be implemented in combination. For example, in the above-described embodiment, the order of the operations and the order of the processes are shown as an example,

and are not limited to these. The same applies to the case where numerical values or mathematical expressions are used in the description of the above-described embodiment.

Claims

1. A method for grinding a workpiece, the method comprising: performing first grinding in which a gear-shaped workpiece and a grinding tool are meshed and rotated, whereby a workpiece tooth surface of the workpiece is ground with a helical grinding tooth surface of the grinding tool; generating first correction information by inverting a first deflection waveform that indicates a relationship between a rotational phase of the workpiece and a positional deviation amount of the workpiece tooth surface with respect to the grinding tooth surface during the first grinding; and performing second grinding in which in a state where the workpiece and the grinding tool are meshed and rotated, a rotational speed of the workpiece or the grinding tool is changed based on the first correction information, whereby the workpiece tooth surface is ground with the grinding tooth surface.
 2. The method for grinding the workpiece according to claim 1, wherein a difference between a command value of the rotational phase of the workpiece and a detected value of the rotational phase of the workpiece during the first grinding is acquired as the positional deviation amount.
 3. The method for grinding a workpiece according to claim 1, further comprising generating second correction information by inverting a second deflection waveform indicating a relationship between a rotational phase of the workpiece and a positional deviation amount of the workpiece tooth surface with respect to the grinding tooth surface during the second grinding, and performing third grinding in which in the state where the workpiece and the grinding tool are meshed and rotated, the rotational speed of the workpiece or the grinding tool is changed based on the second correction information, whereby the workpiece tooth surface is ground with the grinding tooth surface.
 4. A guiding system comprising a controller including one or more processors executing computer-executable instructions stored in memory, wherein the one or more processors execute the computer-executable instructions to cause the controller to: perform first grinding in which a gear-shaped workpiece and a grinding tool are meshed and rotated, whereby a workpiece tooth surface of the workpiece is ground with a helical grinding tooth surface of the grinding tool; generate first correction information by inverting a first deflection waveform that indicates a relationship between a rotational phase of the workpiece and a positional deviation amount of the workpiece tooth surface with respect to the grinding tooth surface during the first grinding; and perform second grinding in which in a state where the workpiece and the grinding tool are meshed and rotated, a rotational speed of the workpiece or the grinding tool is changed based on the first correction information, whereby the workpiece tooth surface is ground with the grinding tooth surface.
 5. The grinding system according to claim 4, wherein the one or more processors cause the controller to acquire, as the positional deviation amount, a difference between a command value of the rotational phase of the workpiece and a detected value of the rotational phase of the workpiece during the first grinding.
 6. The grinding system according to claim 4, wherein the one or more processors cause the controller to: generate second correction information by inverting a second deflection waveform that indicates a relationship between a rotational phase of the workpiece and a positional deviation amount of the workpiece tooth surface with respect to the grinding tooth surface during the second grinding, and perform third grinding in which in the state where the workpiece and the grinding tool are meshed and rotated, the rotational speed of the workpiece or the grinding tool is changed based on the second correction information, whereby the workpiece tooth surface is ground with the grinding tooth surface.
-

